# Painter Naming and the Distributional Gap Between Generated Images and Original Paintings

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**Abstract**

Naming a painter may bring generated images closer to that painter's works without reproducing their distribution. We examine this distinction using 31 interpretable color, spatial and texture features. An exploratory comparison of 1,536 painter-conditioned images with 649 digital reproductions by Monet, Sisley, Pissarro and Cézanne finds lower generated spread and high balanced classification accuracy across all 24 service-alias, painter and prompt combinations. A controlled study then compares 1,006 outputs from three services with a separate 70-work Monet/Cézanne panel. Adding the painter's name to fixed scene prompts lowers energy discrepancy in all six service–painter comparisons; four reject the prespecified conditional null after multiplicity adjustment. Descriptive analyses show improved joint alignment with the two painter panels and reduced spread. At the main neighborhood scale, median reference coverage increases in five comparisons and ties in one, but remains below that of matched held-out paintings. Scene retrieval improves in two comparisons and declines in four, showing that contraction alone does not determine scene distinguishability. A separate 192-image palette experiment estimates the additional chroma-response effects of Monet and Cézanne clauses relative to a generic painting clause as $-0.284$ and $-0.018$ development-IQR units. Both approximate simultaneous 95% intervals include zero ($[-0.585, 0.016]$ and $[-0.343, 0.308]$). A secondary comparison finds a smaller response under the generic clause than under the artist-free control. These results distinguish improved feature proximity from distribution matching, while leaving the perceptual significance and causes of the remaining gap unresolved.

## 1 Introduction

A generated painting can evoke a painter while the generated collection differs systematically from that painter's works. A familiar palette or motif does not establish that the collection reproduces the distribution of composition, color and texture found in paintings. Conversely, a broad output distribution can be far from the intended reference collection. Evaluation therefore requires comparisons of location, spread and the organization of variation, beyond image distance to an average feature vector.

We examine three questions. How do generated feature distributions differ from those of digital reproductions of original paintings? What changes when a painter's name is added to an otherwise fixed scene prompt? Does the accompanying change in spread indicate a weaker response to instructions? Our target is a finite reference panel measured in a fixed feature space. A naming intervention may change depicted content and rendering as well as stylistic cues, so its measured effect concerns the output of the image service as a whole.

An exploratory corpus of Monet, Sisley, Pissarro and Cézanne provides the broad distribution comparison across three prompt formulations. Study 1 then tests painter naming on separate Monet and Cézanne panels, using three services and 24 common scenes with three repetitions each. Energy discrepancy measures reference proximity; comparisons with both painter panels assess relative alignment; spread and reference-neighborhood coverage describe the remaining gap. Repeated scenes distinguish variation between scenes from variation among outputs of the same prompt. A retrieval diagnostic asks whether these scenes become easier or harder to distinguish in feature space.

Study 2 tests a related, narrower hypothesis: whether named painting clauses attenuate the response to an explicit palette instruction beyond the effect of a generic painting clause. It uses six new scenes and 192 outputs from one service. The different scenes and palette manipulation test an observable instruction response without identifying a cause of Study 1's contraction. Table 1 summarizes the cohorts and separates prespecified inference from exploratory and subsequent descriptive analyses.

The contribution is a controlled empirical characterization of painter naming, rather than a new distribution metric. Naming improves primary proximity and often increases reference-neighborhood coverage while reducing spread; these changes do not eliminate the discrepancy from reference paintings. Contraction accompanies both improvements and declines in scene retrieval, and the palette experiment leaves additional named-clause effects unresolved. Together, these findings show why a smaller feature distance alone is insufficient to characterize resemblance to a painter's distribution.

## 2 Related work

Color, image structure and information-theoretic measures have been used to study variation across painting collections (Kim et al., 2014; Sigaki et al., 2018; Lee et al., 2020). Kim et al. (2026) contrasted autoencoder and CLIP representations and investigated context-keyword image-to-image transformations in art-historical development. Our text-only interventions compare generated distributions with fixed painter reference panels, without supplying an original composition as input.

Deliège et al. (2025) combine three experts' assessments with quantitative measures of relative shift, dispersion and overlap for five movements and ten painters. Their generated sample contains 300 Midjourney v6 images, whereas historical reference ranges come from the experts' knowledge of artistic production, without presenting a fixed historical-image corpus. Our references are explicit panels of digital reproductions, and our repeated scene prompts support a within-service naming contrast. The studies therefore use different reference targets and measurements: expert characterizations of style versus observed image features.

AI-Pastiche evaluates artistic imitation through a public authenticity survey and prompt-adherence assessments by the research team and additional volunteers (Asperti et al., 2025). AI-WikiArt conditions generation on captions of original paintings and evaluates attribution with vision-language models (Fu et al., 2025). These studies address imitation and attribution under different conditioning and judgment procedures. Our comparison holds scene text fixed across naming conditions, linking distributional changes to a specified prompt intervention.

Asperti (2026) report human/generated separation in CLIP space and probe it using interpretable descriptors, multiscale representations and image inversion. Their preprint finds that embedding displacement can have low perceptual salience, reinforcing the distinction between measurable image differences and stylistic judgments. Here, the focus is how prompt assignment changes the generated distribution.

Generative-model evaluation distinguishes fidelity, diversity and coverage (Naeem et al., 2020), and preprocessing can materially alter feature comparisons (Parmar et al., 2022). We apply energy discrepancy (Székely and Rizzo, 2013) alongside spread and neighborhood measures. Compositional benchmarks evaluate attribute binding and object relations, including color binding (Huang et al., 2023). Our palette experiment measures global median chroma, a different aspect of instruction response from object-specific color assignment.

## 3 Study overview and feature representation

The analyses share a fixed image representation but use separate cohorts and comparisons (Table 1). Throughout, *reference proximity* means lower energy discrepancy, *spread* means total feature variance, and *contraction* means lower spread under named than artist-free prompting. These quantities describe digital image distributions.

Each image is measured in 31 interpretable coordinates: 11 color, eight spatial and 12 digital-texture features (Appendix A.1). The primary pipeline applies orientation metadata, converts valid embedded color profiles to sRGB, assumes sRGB for compatible unprofiled files and preserves aspect ratio at a 512-pixel short side. Images are opaque and resized with Lanczos filtering, without upsampling or discretionary cropping. Each coordinate is standardized by its median and interquartile range (IQR) in a separate development collection of 221 works: 101 Monet, 36 Sisley, 48 Pissarro and 36 Cézanne, with equal total weight per painter. Reference and generated images are excluded from fitting this transformation. The scaler therefore uses separate works by the same painters, rather than unseen painter identities.

For reference vectors $X = \{x_i\}$ and generated vectors $Y = \{y_j\}$, let $a_i, b_j$ be nonnegative weights summing to one within each set. We use the empirical V-statistic energy discrepancy

$$\widehat{\mathcal{E}}(X,Y) = 2\sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,k} a_i a_k \|x_i - x_k\|_2 - \sum_{j,l} b_j b_l \|y_j - y_l\|_2. \tag{1}$$

This statistic incorporates distances within and between the two collections. Its finite-sample baseline need not be zero for independent samples from the same population. We interpret lower values as closer empirical distributions in the specified feature geometry, not as a calibrated degree of artistic similarity.

## 4 Exploratory comparison across four painters

### 4.1 Corpus and descriptive methods

The exploratory corpus comprises 297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne digital reproductions, selected through recorded outdoor-place metadata. Two requested OAuth aliases, `gpt-image-1` and `gpt-image-2`, each supply 64 images per painter and prompt method: 16 scene templates repeated four times. The aliases are request labels, not independently attested model snapshots. Three formulations name the painter: a by-name prompt; an explicit style instruction appended to the artist-free scene; and that instruction plus attention to color relationships, organization of forms and painted marks. This gives 1,536 painter-conditioned outputs. A further 384 artist-free outputs provide method-specific controls shared across painter comparisons. Appendix A.3 gives the exact construction of the three named/control contrasts. No original painting is supplied to generation.

Two initially refused requests, one for Pissarro and one for Cézanne, succeeded in later trials. These outputs complete a 1,920-image descriptive grid, but their later timing prevents the planned randomization inference for the original full grid. All analyses in this section were developed after the data were observed.

For visualization, each painter has one weighted principal-component analysis (PCA) basis fitted to its originals and all six alias/method groups. Originals carry half the total mass, and each generated group carries one twelfth. Every prompt panel for a painter shares that basis and its limits. We additionally measure the ratio of separately centered sample covariance traces, using denominator $n - 1$ within each group. This descriptive ratio differs from the content-weighted population-form trace used in Study 1.

| Analysis | Data and comparison | Evidential role |
|---|---|---|
| Four-painter exploration | 649 references; 1,536 painter-conditioned and 384 artist-free outputs from 2 requested aliases, 3 prompt methods and 16 repeated scenes | Post-hoc distribution, spread, grouped discrimination and matched-size reference comparisons; separate from the two controlled cohorts. |
| Study 1 primary | 864 detailed outputs: 3 services $\times$ 2 painters $\times$ 24 scenes $\times$ 3 repeats $\times$ 2 conditions; 142 additional short-scene outputs | Eight prespecified paired energy tests: six named/free and two detailed/short-scene named contrasts. |
| Study 1 descriptive | Same images; 38 Monet and 32 Cézanne references | Prospective spread, PCA, detection, coverage and painter-distance summaries; subsequent alignment double contrasts, matched-real coverage, decomposition, retrieval and extended sensitivities are post-result. |
| Study 2 primary | 192 new outputs: 1 service $\times$ 6 fixed scenes $\times$ 4 repeats $\times$ 4 arms $\times$ 2 palettes | Two prespecified named-minus-generic chroma-response interactions; shared free/generic controls and secondary nominal contrasts. |

**Table 1:** Design and evidence hierarchy. Exploratory collection: 5–6 September 2026 UTC, including two later retries. Study 1 collection: 6–7 September 2026 UTC; Study 2: 8 September 2026 UTC. Painter selection followed earlier exploration, while both studies' primary contrasts were fixed before their generation. Sensitivity views reuse images; they are not independent replications.

Fixed linear and radial-basis-function (RBF) kernel-ridge classifiers distinguish generated images from originals in all 31 coordinates, without using PCA scores or selecting hyperparameters. Class weights are balanced and regularization is 0.01. The kernels are $K_{\text{lin}}(x, z) = x^\top z/31 + 1$ and $K_{\text{RBF}}(x, z) = \exp(-\|x - z\|^2/31) + 1$; the constant is a regularized intercept. Sixteen folds hold out one complete generated scene, including its four repeats, and a disjoint set of original work identities. Balanced accuracy averages the correct-classification rates for original and generated images; 0.5 is the chance reference. These splits do not hold out independent capture sources or service sessions.

To assess the finite-sample energy baseline, 128 deterministic splits divide each painter's originals into disjoint groups of size $n = \min(64, \lfloor N_p/2 \rfloor)$. Generated/reference comparisons use the same reference subsets and feature-independent generated subsets of size $n$. Each reported generated/reference median summarizes 768 distance estimates: 128 from each of six alias/method groups. Images are not pooled across groups. The same works recur across draws, so the resulting medians describe subsampling sensitivity rather than independent replications.

### 4.2 Distribution differences across all four painters

Figure 1 shows overlapping clouds with different locations and concentrations across all three formulations. The displayed PCs retain 44.1%, 44.4%, 47.8% and 42.9% of balanced variation for Monet, Sisley, Pissarro and Cézanne, respectively. More than half the variation lies outside these planes. The classifiers in Table 2 assess distinguishability in all 31 coordinates, including the variation omitted from the plots.

All four painters show lower generated spread and high classification accuracy (Table 2). The generated/reference trace ratios range from .206 to .376, and RBF balanced accuracy ranges from .940 to .983. The magnitude varies across painters and prompt formulations, while the direction of the spread difference is consistent across all 24 groups.

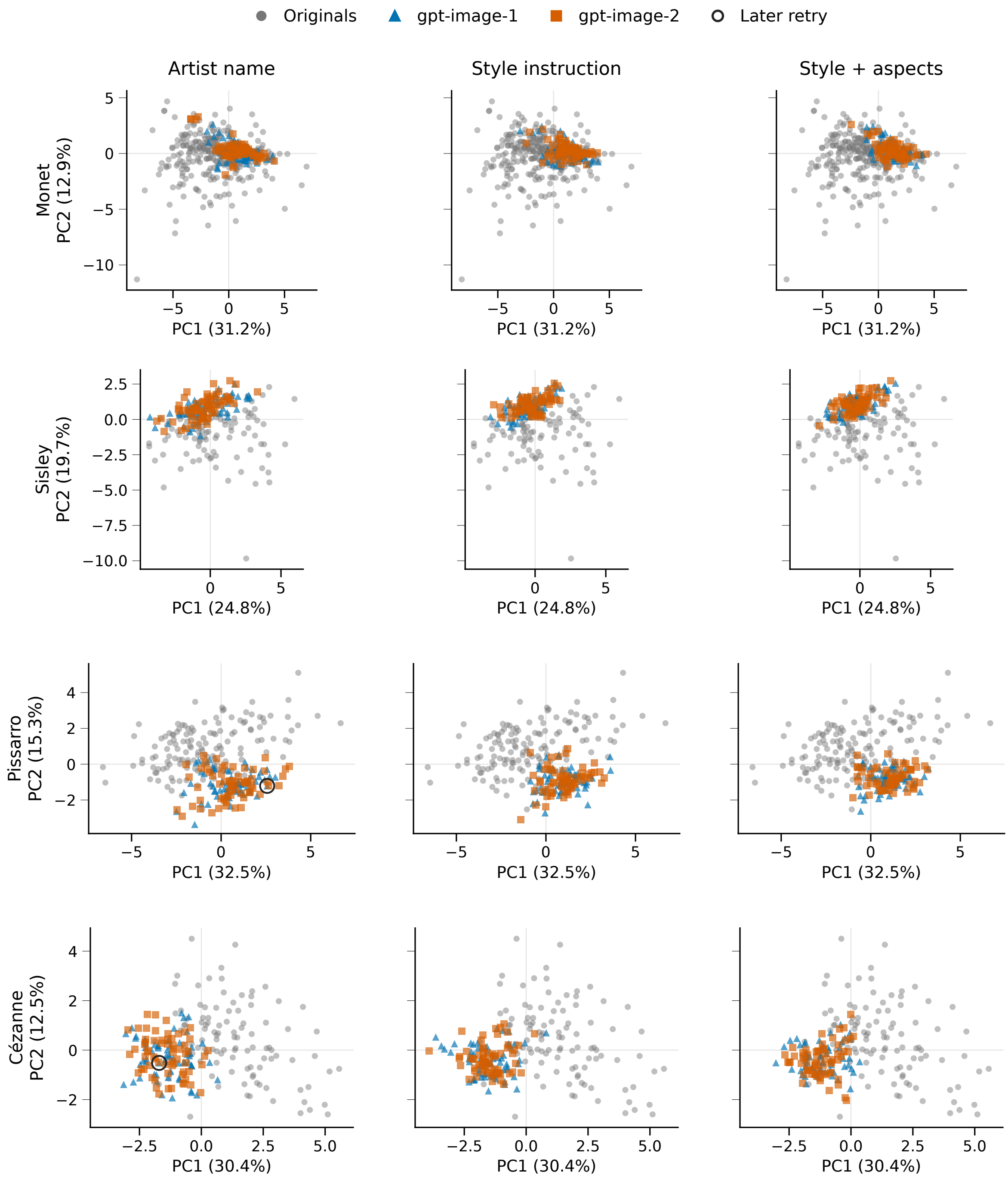


**Figure 1:** Original and generated distributions for all four painters and all three prompt methods. Gray points are originals; blue triangles and orange squares are the two requested service aliases. Each cell contains 64 outputs per alias. Black rings identify the two later retries. Originals are repeated across columns for comparison. All points and outliers are retained, with equal geometric aspect and common limits within each painter. Each row uses its own PCA basis; directions and coordinates cannot be compared across painters. The projection is descriptive and is not used by the classifiers.

| Painter | References | Trace ratio | Linear BA | RBF BA |
|---|---|---|---|---|
| Monet | 297 | .206–.304 | .946–.973 | .947–.982 |
| Sisley | 106 | .219–.353 | .947–.976 | .970–.983 |
| Pissarro | 141 | .240–.376 | .934–.950 | .940–.979 |
| Cézanne | 105 | .249–.309 | .913–.964 | .942–.978 |

**Table 2:** Four-painter distribution summaries in all 31 features. Ranges cover all six alias/method cells per painter, each with 64 generated images; they are not uncertainty intervals. Trace ratios compare generated with original sample variance. Both fixed classifiers use held-scene/disjoint-work validation. Appendix B reports every cell.

| Painter | Matched $n$ | Original/original | Named/original | Free/original |
|---|---|---|---|---|
| Monet | 64 | .1932 | 2.0683 | 1.7319 |
| Sisley | 53 | .2149 | 1.6776 | 1.8779 |
| Pissarro | 64 | .1635 | 1.9745 | 1.8537 |
| Cézanne | 52 | .1989 | 2.1419 | 1.6570 |

**Table 3:** Median all-feature energy discrepancy in matched-size subsets. Original/original medians summarize 128 disjoint-within-draw splits; each generated median summarizes $6 \times 128$ distance estimates. Works recur across draws; the medians are descriptive baselines, not equivalence thresholds or comparisons of independent capture sources.

Both generated/reference medians exceed the original/original median for every painter (Table 3), so unequal group sizes alone do not explain these energy gaps. The named median is lower than the artist-free median only for Sisley. In the separate feature-family prompt comparisons, by-name prompts have lower energy than explicit style instructions in all 24 alias–painter–family comparisons; adding aspect instructions reduces it in 13 of 24. These directional counts summarize correlated descriptive comparisons. Artist-free outputs also have high RBF balanced accuracy (.915–.990) and generated/reference trace ratios .369–.750, so discrimination does not isolate painter-specific mismatch.

The measured gap is compatible with content and capture differences as well as generation behavior. Equalizing four coarse content categories leaves named population-trace ratios at .211–.372, but categories describe recorded works and requested scenes, not independently verified content matches. The exploratory comparison therefore identifies differences among the observed image collections. Study 1 tests a narrower prompt intervention with a separate reference panel and common scene descriptions.

# 5 Study 1: painter naming and distributional proximity

## 5.1 Design and reference panels

We compare 38 Monet and 32 Cézanne works represented by Wikimedia-delivered digital reproductions. Monet and Cézanne were selected after the four-painter exploration to prioritize contrasting observed patterns, additional service families and reference curation within a fixed collection budget. The reference works and generated images form a separate cohort; neither is pooled with the exploration. Reference identities were screened against prior project exposure. Availability and landscape eligibility determine this selected panel, rather than probability sampling of an oeuvre. Three coarse classes have water/built/land counts of 21/4/13 for Monet and 3/11/18 for Cézanne. A maintainer-run LLM coded these classes without independent human adjudication; 17 mixed or uncertain annotations remain flagged.

All detailed prompts begin "Create an oil painting on canvas." Naming adds "In the style of [painter]." before the otherwise identical scene and closing instructions. Eight scenes per class each have three repeats. No original image is supplied. The short-scene named condition on one service changes description specificity as well as length; both versions name the painter.

We request Nano Banana 2 (NB2; `gemini-3.1-flash-image`) and FLUX (`flux.2-max`) through fixed OpenRouter providers, and `gpt-image-2` through a local OAuth service. These identify requested services, not immutable weights. Conditions are randomly ordered within each service/painter/scene/repetition group. Collection spans 11.5 hours on 6–7 September 2026 UTC. The 1,008 assigned slots yield 1,006 images; two Cézanne short-scene requests are refused. All successful outputs are retained. Returned geometry and quality are part of the service outcome: all 576 NB2/FLUX outputs are square, whereas all references are nonsquare; OAuth rendering varies by condition. Appendix A gives prompts, delivery details and missingness accounting. Generated classes denote requested content, not verified adherence.

### 5.2 Analysis

**Weighting and spread.** References have equal weight, $a_i = 1/N_p$. If painter $p$ has $N_{pc}$ reference works in class $c$ and a generated cell has $n_{pc}$ available images in that class, a generated image receives $b_j = N_{pc}/(N_p n_{pc})$. Thus both distributions have the reference panel's class proportions. This weighting reduces differences in broad content mixture but leaves within-class content differences unresolved.

To measure spread, we use the weighted population-form covariance trace

$$\mathcal{V}(Y) = \sum_j b_j \|y_j - \bar{y}\|_2^2, \qquad \bar{y} = \sum_j b_j y_j. \tag{2}$$

We distinguish the generated/reference ratio $\mathcal{V}(Y)/\mathcal{V}(X)$ from the named/artist-free ratio $\mathcal{V}(Y_{\mathrm{N}})/\mathcal{V}(Y_{\mathrm{F}})$. The first measures spread relative to paintings; the second measures the change under naming. Sums of squared coordinate-wise IQRs provide a complementary summary.

Repeated descriptions allow an exact descriptive decomposition. For description $b$, let $w_b$ be its total image weight and $\bar{y}_b$ its weighted mean. Then

$$\mathcal{V}(Y) = \underbrace{\sum_b \sum_{j \in b} b_j \|y_j - \bar{y}_b\|_2^2}_{W\text{: within descriptions}} + \underbrace{\sum_b w_b \|\bar{y}_b - \bar{y}\|_2^2}_{B\text{: between description means}}. \tag{3}$$

The between-description term can be further separated into within-class and between-class components. These are exact summaries of the observed images. With only three repetitions per scene, the between-description component also contains generation noise in the estimated scene means; it is not a bias-corrected population variance component.

**Primary contrasts and inference.** The prespecified inferential family comprises six named-minus-artist-free energy contrasts and two detailed-minus-short-scene named contrasts on OAuth. A contrast uses complete pairs sharing the description, repetition and assigned batch, with the same content weights in both conditions. The paired statistic retains all within- and between-collection terms in Equation 1. Appendix C gives the swap identity.

For each contrast, 99,999 Monte Carlo sign assignments yield a two-sided randomization $p$-value with the plus-one correction. Holm adjustment controls the fixed family of eight tests at level .05. The sharp null states that prompt assignment changes neither availability nor measured features in any assigned position, under the fixed request and technical-retry policy and no interference. For a pairwise OAuth comparison, the third condition's position is conditioned on. Inference is conditional on the assigned prompts and reference panel; it does not estimate an average effect over a painter's oeuvre or over new scenes. Treatment-dependent call duration or service carryover could violate no interference.

Study 1's prompt inventory, primary metrics and eight-test family were specified before its generation. Its original spread, coverage, grouped detection, PCA and own/cross-painter distance summaries were also prospectively specified, as descriptive analyses. Later feature-view and reference sensitivities, the alignment double contrast, matched-real coverage, variance decomposition, retrieval and cross-service detection are post-result diagnostics. They introduce no additional confirmatory tests.

**Alignment and distribution diagnostics.** A relative alignment contrast compares both painter-conditioned collections $G_M, G_C$ with both reference panels $X_M, X_C$:

$$I = \widehat{\mathcal{E}}(X_M, G_M) + \widehat{\mathcal{E}}(X_C, G_C) - \widehat{\mathcal{E}}(X_M, G_C) - \widehat{\mathcal{E}}(X_C, G_M). \tag{4}$$

All four distributions use equal one-third content-class masses, so panel mixture differences cannot by themselves produce the interaction. Within-distribution energy terms cancel; negative $I$ favors joint own-painter pairing. Identical artist-free payloads under the two painter allocation labels provide a control for collection variation. This joint statistic does not require each named collection to be individually nearest its own panel.

We use the coverage construction of Naeem et al. (2020): the fraction of reference neighborhoods containing at least one query image. Matched generated queries and disjoint real queries are evaluated against the same reference anchors. Each of 100 fixed draws matches query counts within content class; neighborhood radii use the first, third or fifth nearest other anchor. The resulting summaries describe the fixed panel (Appendix D.1). Processing and representation checks cross three pipelines with four feature views; reference deletions and four development-painter omissions assess influence (Appendix D.3).

**Scene distinguishability under contraction.** Using the Study 1 detailed named and artist-free conditions, we classify each held-out repetition by its nearest scene centroid in the fixed feature space. For each service, painter and condition, centroids use the other two repetitions of each scene; the held-out image never enters its own or any competing centroid. Cycling through the three repetitions yields 72 queries per condition. We evaluate both all 24 scenes and the eight scenes in the query's broad content class. The latter comparison removes discrimination attributable solely to broad class. The scaler remains fixed on development paintings.

Retrieval uses Euclidean distances in all 31 coordinates, color alone, spatial features alone, texture alone and the 19 non-texture coordinates, under each processing pipeline. Every scene has equal weight. For this diagnostic, variance components are also recomputed with equal scene weights; these differ from the reference-class-weighted components in Equation 3. The comparisons are descriptive: queries have overlapping centroid-training sets and are not 72 independent Bernoulli trials. Under a uniform transformation $y' = ay + t$, $a > 0$, within- and between-scene traces both scale by $a^2$, while all retrieval assignments are unchanged. Retrieval measures relative scene distinguishability, not absolute response gain or the presence of requested objects.

### 5.3 Painter naming improves primary feature proximity

Painter naming lowers primary energy discrepancy in all six service–painter comparisons (Figure 2). NB2 decreases from 4.188 to 3.342 for Monet and from 4.704 to 2.886 for Cézanne; FLUX decreases from 1.990 to 1.365 and from 1.700 to 0.804, respectively. These four contrasts reject the conditional sharp null after adjustment across the eight-test family (Table 4). The smaller OAuth differences do not reject the null after adjustment, nor do the detailed-versus-short-scene comparisons. Non-rejection does not establish equivalence.

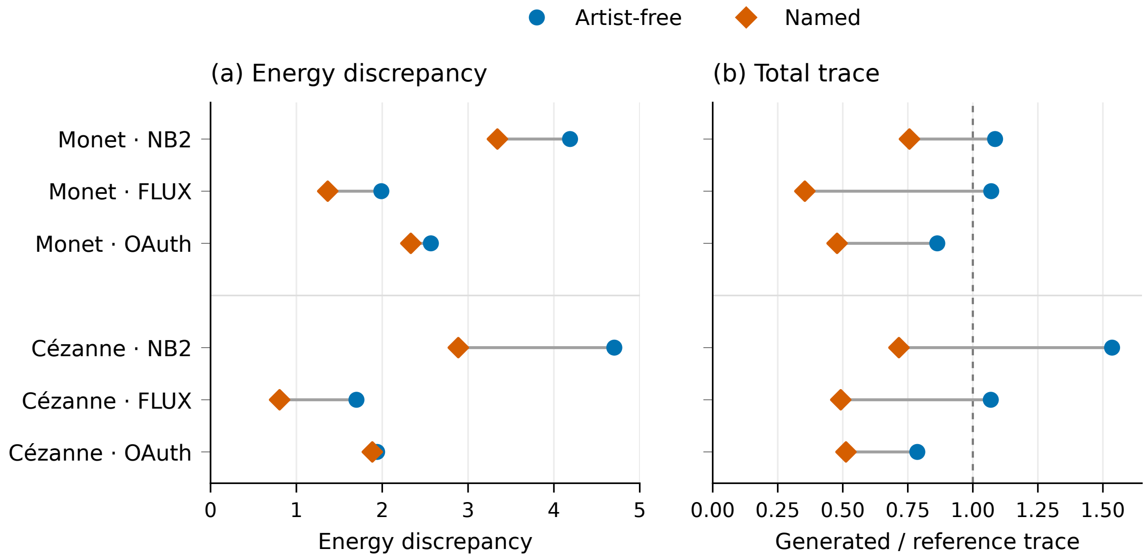


**Figure 2:** Primary feature discrepancy and spread under artist-free and named prompting. Energy compares each generated distribution with its painter's reference panel. Trace ratios compare generated spread with reference spread; the line at one denotes equal empirical trace. Each displayed generated cell contains 72 images; reference panels contain 38 Monet and 32 Cézanne works. The panels are descriptive and do not share a numerical scale.

| Comparison | Service | Painter | Pairs | $\Delta\widehat{\mathcal{E}}$ | Holm $p$ |
|---|---|---|---|---|---|
| Named − free | NB2 | Monet | 72 | −0.846 | .00310 |
| | NB2 | Cézanne | 72 | −1.818 | .00008 |
| | FLUX | Monet | 72 | −0.625 | .00008 |
| | FLUX | Cézanne | 72 | −0.896 | .00008 |
| | OAuth | Monet | 72 | −0.234 | .19308 |
| | OAuth | Cézanne | 72 | −0.056 | 1.00000 |
| Detailed − short scene | OAuth | Monet | 72 | −0.037 | 1.00000 |
| | OAuth | Cézanne | 70 | −0.272 | .09280 |

**Table 4:** The complete prespecified test family, using the primary 31-feature representation and matched pairs. Negative changes indicate smaller energy in the first condition. Values are conditional randomization results, with 99,999 sign draws and Holm correction across all eight comparisons. Both conditions in the final two rows name the painter.

The same primary feature space also shows that the named collections have less aggregate spread than the references. Generated/reference trace ratios for the six detailed named cells range from .354 to .756, whereas the four NB2 and FLUX artist-free cells range from 1.069 to 1.535. Thus, reduced spread is not a general property of all generated images in this experiment. It is associated with the tested painter-naming conditions. The two short-scene named OAuth cells have still lower reference-relative ratios, .306 for Monet and .395 for Cézanne, although their energy contrasts do not establish an advantage of detailed prompting.

Energy and spread are distinct but dependent summaries. For FLUX/Monet, twice the cross-domain mean distance falls from 15.831 to 12.421, while the within-generated mean distance falls from 6.906 to 4.120. Because the latter is subtracted in Equation 1, its decrease opposes rather than mechanically produces the observed energy improvement. The cross-domain change is larger. The coexistence of lower discrepancy and contraction is an empirical result here, not a necessary fidelity/diversity trade-off.

### 5.4 Relative painter alignment and reference coverage

Equal-class comparisons address whether the two names merely produce a common movement toward generic paintings. Naming makes the joint own/cross-painter interaction more negative on all three services (Table 5). The named-minus-free change remains negative across all 36 service/view/pipeline combinations. This joint relative contrast does not require every generated collection to be closest to its intended painter panel. In the primary equal-class comparison, NB2/Monet remains closer to the Cézanne panel (energy 3.488) than to Monet (3.687), despite improved joint alignment across the two named collections.

| Service | Artist-free $I$ | Named $I$ | Named $-$ free |
|---|---|---|---|
| NB2 | .101 | $-$.855 | $-$.956 |
| FLUX | $-$.048 | $-$1.755 | $-$1.707 |
| OAuth | $-$.044 | $-$1.802 | $-$1.758 |

**Table 5:** Joint painter alignment under equal content-class weights in the primary feature space. Negative $I$ favors own-painter over cross-painter pairing (Equation 4). Separately generated artist-free collections have identical prompt payloads across painter allocation labels. No additional significance test or equivalence threshold is assigned to this diagnostic.

Naming also changes coverage of the reference neighborhoods (Table 6). At $k = 3$, median coverage increases in five service–painter comparisons and ties in OAuth/Monet. It nevertheless remains below the matched real-query median in every named condition. Thus contraction can accompany greater occupation of reference neighborhoods, even while coverage remains incomplete. These are descriptive comparisons using the same reference anchors and class-matched query counts; they introduce no additional significance tests.

| Painter | Service | Artist-free | Named | Held-out real |
|---|---|---|---|---|
| Monet | NB2 | 0.278 | 0.500 | 0.944 |
| Monet | FLUX | 0.556 | 0.833 | 0.944 |
| Monet | OAuth | 0.500 | 0.500 | 0.944 |
| Cézanne | NB2 | 0.200 | 0.400 | 0.933 |
| Cézanne | FLUX | 0.467 | 0.800 | 0.933 |
| Cézanne | OAuth | 0.467 | 0.533 | 0.933 |

**Table 6:** Median reference-neighborhood coverage at $k = 3$ across 100 matched finite-panel draws. Each draw uses the same original anchors for real and generated queries, with 18 queries for Monet and 15 for Cézanne. Generated query counts and content classes match the real queries. The medians describe the reused panel; they are not population confidence intervals.

Coverage depends on neighborhood size. At $k = 5$, FLUX/Cézanne reaches 1.000 for both named and real queries (Appendix D.1). This sensitivity precludes treating a coverage value as evidence of distribution equality.

### 5.5 Variation within and between scenes

The named/artist-free total-trace ratios range from .331 to .697 in the primary representation. Decomposition reveals different sources of this contraction (Figure 3). FLUX reduces both within-description and between-description variation strongly: the respective ratios are .309 and .337 for Monet, and .489 and .451 for Cézanne. For NB2/Cézanne, within-description variation decreases to .786 while between-description variation decreases to .378. Its aggregate reduction therefore conceals a much larger relative change between descriptions than between repetitions of a description.

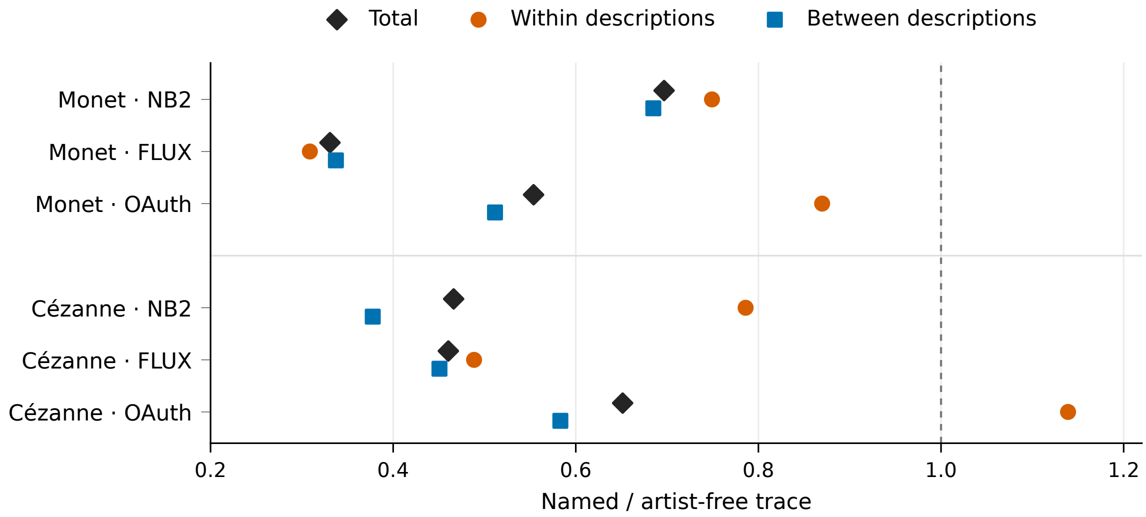


**Figure 3:** Where generated variation changes under naming. Each point is a named/artist-free trace ratio in the primary representation. Total variation is the sum of within-description and between-description components, so its ratio need not equal their unweighted average. The line at one denotes no change. OAuth/Cézanne contracts in total while increasing within-description variation. These finite-set decompositions are descriptive.

The OAuth/Cézanne case shows why the distinction is necessary. Total trace falls to .651 of its artist-free value even though within-description trace rises to 1.139; between-description trace falls to .583. A tighter overall cloud can therefore coexist with more variable repetitions. The decomposition thus locates changes that an aggregate spread measure conceals.

### 5.6 Contraction need not reduce scene distinguishability

In the primary 31-feature retrieval diagnostic, naming improves accuracy in two of six service–painter cells and lowers it in four (Figure 4). FLUX/Monet improves from 44.4% to 59.7%, although its equal-scene between-description trace falls to .336 of the artist-free value. Its retrieval gain remains positive under all three pipelines, including within-class candidate restriction. This example shows that lower between-scene spread can coexist with better retrieval in the observed collection. Transfer to new scenes or service runs remains untested.

The changes are heterogeneous. For Cézanne, accuracy declines from 66.7% to 41.7% on NB2 and from 83.3% to 69.4% on OAuth. Both declines persist across all five feature views and three pipelines in both candidate settings. For Monet, the OAuth decline is smaller within class (84.7% to 83.3%) than across all 24 scenes (79.2% to 70.8%). The NB2 difference is one correct query out of 72. Changes in separability therefore vary across services and painters.

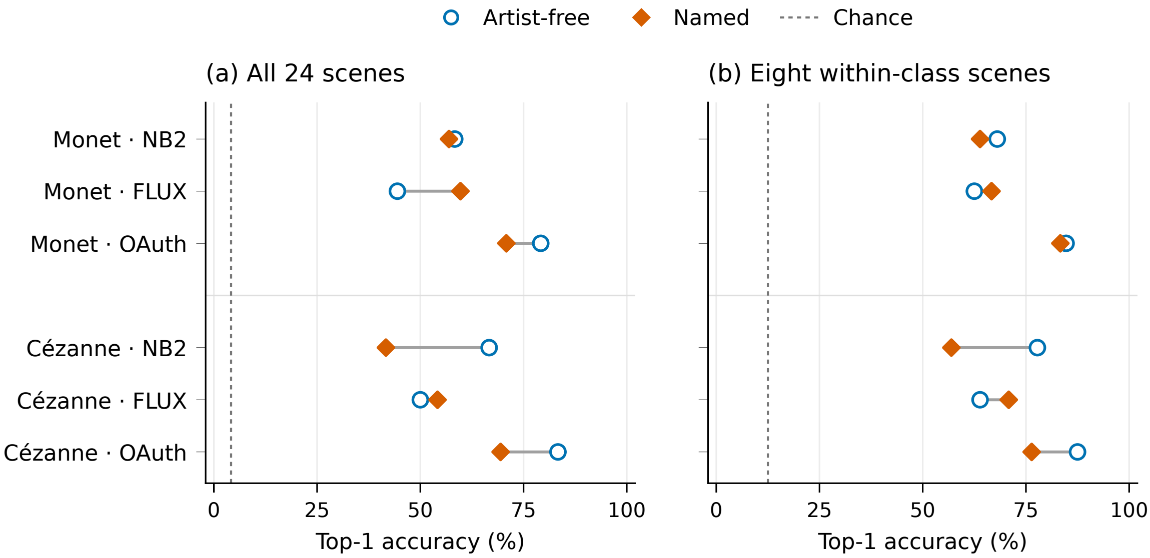


**Figure 4:** Held-repetition scene retrieval in Study 1. Each condition has 72 queries; centroids use only the other two repetitions. Panels compare 24 scene candidates with eight candidates in the same broad class. All scenes have equal query weight. Dashed lines mark chance rates of 1/24 and 1/8, respectively. These are descriptive accuracies in the primary 31-feature representation, not semantic-adherence scores or independent scene-population estimates.

### 5.7 Sensitivity to representation and references

All 72 named/artist-free total-trace ratios across the four feature views and three pipelines are below one, ranging from .308 to .816. Proximity is less consistent: 66 of 72 energy changes are negative. Both FLUX painter comparisons and NB2/Cézanne remain negative in all 12 view/pipeline combinations. For NB2/Monet, the 256-pixel color/spatial-only comparison instead increases energy by .416. The same no-texture view at 512 pixels gives a named/reference trace ratio of 1.101. Consequently, robust contraction relative to artist-free outputs does not imply robust below-reference spread or uniformly improved proximity.

The representation places substantial weight on texture. Texture coordinates contribute 49.0% and 47.1% of primary reference trace for Monet and Cézanne. Some multiscale coordinates are strongly correlated in development data. The feature views assess dependence on this representation; they reuse the same images and do not provide independent evidence.

Deleting individual reference works, holding-collection groups or content classes preserves the direction of all four NB2/FLUX primary proximity changes. These directions also survive omission of any one development painter from scaling. Some OAuth changes cross zero under these perturbations. These checks show stability of the NB2/FLUX directions within the available panel, while leaving shared reproduction differences unresolved.

## 6 Study 2: response to palette instructions

### 6.1 Design

Study 2 crosses six fixed landscape descriptions (two per broad class), four style arms, two palette instructions and four repetitions, yielding 192 images. All arms begin with the same oil-painting request. The artist-free arm adds no style clause; the generic arm adds “In a traditional landscape-painting style.”; the named arms add the Monet or Cézanne clause. The generic clause controls for the addition of painting-style language. Because it is a single alternative phrasing, the comparison cannot isolate artist identity from all other differences in wording. The palette instruction requests either "a muted, low-chroma palette throughout the painting" or "a vivid, high-chroma palette throughout the painting." Scene text and the closing instruction are otherwise identical. The free and generic outputs are shared controls for both painters, giving 48 images per arm.

The 24 scene/repetition blocks were randomly ordered, with all eight arm/palette requests jointly randomized within each block before generation. Collection lasted 49.5 minutes on 8 September 2026 UTC, using the same OAuth service identifier as Study 1. Requests specified one opaque 1024-by-1024 PNG at medium quality. All 192 images were retained and measured under all three processing pipelines. Actual delivery differed from the requested settings: all outputs were nonsquare RGB PNGs, and 174 reported low quality and 18 medium despite a medium-quality request. The medium-quality counts were 10, 4, 3 and 1 in the free, generic, Monet and Cézanne arms. No output was excluded or adjusted on the basis of these delivery fields. The estimated effect therefore includes changes in rendering associated with prompt assignment. Appendix E accounts for an earlier incomplete collection, which is excluded from primary inference.

### 6.2 Endpoint and inference

The prespecified endpoint is median pixel CIELAB chroma in the primary 512-pixel pipeline. Its development-standardized value is

$$C_{\text{img}} = \text{median}_{\text{pixels}} \sqrt{(a^*)^2 + (b^*)^2}, \qquad z = \frac{C_{\text{img}} - 11.24253}{7.34764}.$$

The center and IQR are fixed on the development collection. All processing variants use this same scale, allowing direct comparison in common units. Let $d(a)$ be the equal-scene mean vivid-minus-muted response for arm $a$. The two primary interactions are

$$\kappa_M = d(M) - d(G), \qquad \kappa_C = d(C) - d(G), \tag{5}$$

where $G$ denotes the generic arm and $M, C$ the named arms. Four repeat-block contrasts per scene estimate each interaction; the two estimates share the same generic draws. Template-stratified covariance and Welch–Satterthwaite $t$ approximations provide Bonferroni simultaneous 95% family intervals and Holm-adjusted two-sided tests for the two primary interactions. These model-based inferences condition on six fixed scenes and assume independent repeat-block errors with stable within-scene response distributions. Randomized request order alone does not make the interaction tests exact. Full formulas appear in Appendix E.

Control responses, named-minus-free and generic-minus-free contrasts use nominal 95% intervals as secondary summaries. A negative interaction indicates attenuation of a positive chroma response only if the control response is positive and the named response does not reverse sign. Processing variants and the remaining coordinates are descriptive. The two primary interactions were specified before Study 2 generation and remain separate from Study 1's eight-test family. The allocation is estimation-focused, with no validated minimum relevant effect or power target. A prospective simulation checked the interval/test procedure in three independent-noise proxy scenarios; Appendix E reports its calibration and limits.

To connect responsiveness with original paintings, we compare each generated chroma distribution with the already exposed reference panel using one-dimensional Wasserstein distance, empirical quantiles and both directions of central-range occupancy. Quantiles select the first value whose exact rational cumulative weight reaches the specified probability. We report empirical and equal-class reference weights; generated distributions have equal class mass and, for the combined muted/vivid condition, equal polarity mass. The latter is a designed mixture, not a natural output distribution. These descriptive comparisons locate the manipulated chroma distributions relative to the existing reference panel.

### 6.3 Primary interactions and secondary control responses

Both prespecified painter-minus-generic interaction intervals include zero (Figure 5). The Monet estimate is $\hat{\kappa}_M = -0.284$ (simultaneous 95% interval $[-0.585, 0.016]$, Holm $p = .0649$); the Cézanne estimate is $\hat{\kappa}_C = -0.018$ ($[-0.343, 0.308]$, $p = .8886$). Monet's six scene-specific estimates are all negative, whereas Cézanne's have three positive and three negative signs. Effective Welch degrees of freedom are 15.25 and 9.91, with standard errors .121 and .123, respectively. The 192 images therefore do not constitute 192 independent interaction observations. Appendix E.1 displays all 24 repeat-block interactions per painter, making the variation behind the scene averages visible.

The intervals leave uncertainty about the size and direction of an additional naming effect. Monet's interval allows a reduction of .585 IQR units, about 22% of the generic point response of 2.636. This percentage provides scale context; it is neither a ratio confidence interval nor a validated threshold of practical importance. Cézanne's interval permits decreases and increases of about .3 units. Neither result establishes equivalence with the generic arm.

All four arms respond positively to the vivid rather than muted instruction. The mean responses are 3.489, 2.636, 2.351 and 2.618 development-IQR units for the free, generic, Monet and Cézanne arms, respectively, with nominal 95% intervals above zero. In the secondary comparisons, the generic-minus-free contrast is $-0.853$ (nominal interval $[-1.136, -0.570]$), a 24.5% reduction relative to the artist-free point estimate. Named-minus-free contrasts are $-1.137$ for Monet and $-0.871$ for Cézanne, with nominal intervals $[-1.375, -0.900]$ and $[-1.062, -0.679]$. Thus a smaller response than the artist-free control also occurs under the generic clause; the named/free comparison alone cannot attribute that pattern to painter naming.

The Monet estimate is similar at 256 pixels ($-.301$) and after JPEG90 processing ($-.308$). The JPEG interval narrowly excludes zero, with upper endpoint $-.004$, whereas the primary and 256-pixel intervals have upper endpoints .016 and .003. The inference therefore depends on processing near the rejection boundary; the prespecified primary result remains the basis for

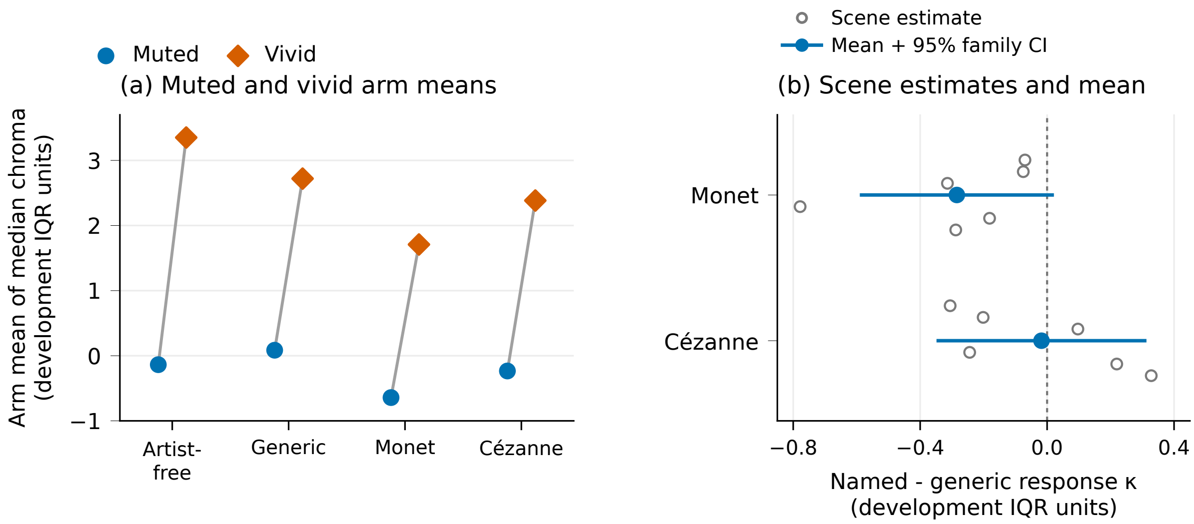


**Figure 5:** Study 2 color levels and additional painter-name interactions. Left: equal-scene mean image-median chroma, with 24 images per arm/polarity; these points have no uncertainty bars. Right: named-minus-generic differences in the vivid-minus-muted response. Gray hollow circles show all six scene estimates per painter; vertical offsets only prevent overlap. Blue points and bars show the equal-scene estimate and approximate simultaneous 95% family interval (Bonferroni marginal coverage 97.5%). Both panels use the same fixed chroma IQR unit but display different estimands. The two interactions share generic draws.

the conclusion. Cézanne remains unresolved under both variants. The arm means also separate color level from responsiveness: relative to generic language, Cézanne lowers the muted and vivid levels by .319 and .337 IQR units, respectively, while leaving their average contrast nearly unchanged. Two levels cannot identify whether such patterns arise from a shifted operating range, response gain or saturation.

The palette manipulation also creates an equal mixture of muted and vivid outputs. Its named/reference Wasserstein distances are lower than those of the generic mixture for both painters, but central-range overlap does not imply equal mass allocation (Appendix F). Because the two-polarity mixture is deliberately constructed, it is an illustration of distribution comparison rather than an explanation of the gap under Study 1's ordinary prompts.

# 7 Discussion

## 7.1 Proximity, distribution matching and instruction response

The results distinguish movement toward a reference distribution from matching that distribution. Across the four-painter exploration, generated collections have lower feature spread than originals and remain distinguishable in the full feature space despite overlap in projection. Matched-size reference comparisons show that unequal counts alone do not explain the energy gaps. In Study 1, naming lowers primary energy discrepancy and improves joint relative alignment with the two painter panels. Median coverage at $k = 3$ also increases in five comparisons and ties in one, although all named medians remain below the matched real medians. The descriptive alignment contrast has no dedicated uncertainty assessment and allows individual cross-painter preferences, as the NB2/Monet example demonstrates.

The naming results also depend on the comparison being made. In the exploratory corpus, the pooled named energy median is lower than the artist-free median only for Sisley. Study 1 finds lower primary energy in all six Monet/Cézanne comparisons, with adjusted rejections in four. The cohorts differ in references, prompt construction, weighting and service conditions. The controlled findings therefore do not establish the same naming effect for Sisley and Pissarro, or supersede the earlier descriptive ordering. Selecting the two follow-up painters after exploration further limits generalization across artists.

Contraction is similarly incomplete as an account of generated variation. OAuth/Cézanne has greater variation among repetitions of the same scene despite lower total spread. FLUX/Monet has lower variation between scene means while retrieval improves. A single aggregate variance ratio conceals these different patterns, and uniform rescaling would leave retrieval unchanged. Evaluation should therefore relate spread to the structure of the prompt design rather than interpreting every reduction as diminished scene distinguishability.

The palette experiment illustrates why the choice of control wording matters. Both primary named-minus-generic intervals include zero, while a secondary comparison finds attenuation under the tested generic clause relative to the artist-free control. That clause is itself an aesthetic instruction. The study thus tests specific wording contrasts, leaving additional painter-name effects uncertain. Its new scenes, explicit palette extremes and single service also prevent attribution of Study 1's contraction to the measured chroma response. Together, the studies support complementary reporting of reference proximity, relative painter alignment, coverage and instruction response, extending corpus-level style comparisons (Deliège et al., 2025) to repeated prompt designs.

## 7.2 Limitations and further research

The feature representation, reference selection and image delivery constrain interpretation. The selected reference panels have unknown capture workflows; broad LLM-assigned classes leave fine composition and generated adherence unmatched. Texture contributes nearly half of Study 1 reference trace, and the NB2/Monet reversal without texture limits conclusions across representations. All NB2/FLUX outputs are square and all Study 1 references are nonsquare, making these domains separable without artistic analysis. OAuth geometry and reported quality also vary by condition. The prompt comparisons include these rendering changes, rather than estimating effects at fixed rendering settings. Human style judgments, independent reference validation and learned-feature comparisons remain unperformed, so the measured gap has no established perceptual interpretation.

Study 2's precision and transfer are limited by six fixed scenes, four repetitions and one collection lasting 49.5 minutes. Its intervals assume stable, independent repeat-block errors; randomization and proxy simulations do not verify those assumptions for the delivered service. The results neither confirm nor exclude additional painter-specific attenuation on new scenes or services.

A prospective extension could vary intermediate palette levels and several generic phrasings to distinguish shifts in color level from changes in response slope or saturation. New scenes and services would test transfer. Matched reproductions and independent style judgments would address whether feature differences track artistic resemblance. More repetitions alone would improve precision without resolving those questions of measurement and control.

## 8 Conclusion

The exploratory collections for Monet, Sisley, Pissarro and Cézanne show lower generated spread and detectable differences from their digital reference panels. In the separate Monet/Cézanne study, naming improves primary proximity and joint relative painter alignment while reducing generated spread. Coverage generally increases at the main neighborhood scale, but the remaining gaps and heterogeneous changes in scene retrieval show why these improvements do not establish distribution matching. The palette experiment leaves additional named-clause effects unresolved and identifies attenuation under the tested generic clause in a secondary comparison. The findings characterize how the measured gap depends on prompts; they do not yet explain its perceptual significance or its origin inside the generation model.

## Data and code availability

The project repository is https://github.com/isingmodel/latent-art-bench. The numerical analyses use local scientific snapshot `28a9eb6`; public archival release is pending. The manuscript source and build scripts are retained with this revision in `paper/`; they postdate the scientific snapshot and are needed to reproduce the current presentation. The snapshot contains prompts, reference identities and annotations, standardized vectors, fixed analysis memberships, code and numerical reports for the four-painter exploration, both controlled studies and the quantile correction. Figure rebuilding uses compact numerical inputs. A separate numeric replay reproduces Study 2's primary estimates and intervals from the recorded chroma values and request schedule, without opening images or responses. This checks calculation from retained measurements; full integrity-checked analysis replay additionally requires the retained raw service-response archive; image remeasurement requires the original and generated pixels. Raw media are not publicly redistributed, and external access has not been arranged. The build guide `paper/README.md` lists per-run inputs and offline commands; Appendix G identifies the evidence locations.

# A Features, prompts and collection details

## A.1 Feature inventory

| Family | Count | Coordinates |
|---|---|---|
| Color | 11 | Lightness median/IQR; chroma median/IQR; chromatic fraction; hue concentration/entropy; chromatic differences at three scales and their slope. |
| Spatial | 8 | Spectral slope/residual/anisotropy; orientation entropy; horizontal–vertical balance; gradient median/IQR; quadrant divergence. |
| Digital texture | 12 | Four wavelet energies, their slope and curvature; three local-binary-pattern entropies; local coefficients of variation at three scales. |

**Table 7:** The fixed interpretable representation. Coordinates describe digital image statistics, not physical paint relief. The 28-coordinate view removes chromatic-difference slope, wavelet slope and wavelet curvature.

Color coordinates use CIELAB with D65 illumination and the 2° observer. Chromatic pixels satisfy $C^* \geq 5$; chroma-weighted hue concentration and 24-bin normalized hue entropy are set to zero when chromatic pixels occupy less than 1% of the image. Chromatic differences are median horizontal/vertical CIEDE2000 distances at lags of 1%, 4% and 16% of the short side; their summary slope is fitted in log–log coordinates.

Spatial and texture coordinates use linear-sRGB luminance. Spectral summaries use a Tukey window with parameter .1 and 32 logarithmic radial bins spanning 4–128 cycles per short side; the spectral slope uses a Theil–Sen fit. Gradient summaries use Scharr derivatives, an 18-bin magnitude-weighted orientation histogram and mean quadrant Jensen–Shannon divergence from the whole-image histogram. Texture coordinates comprise log mean squared detail energies at four levels of an undecimated `db2` wavelet transform, their slope and curvature, normalized uniform-LBP entropies at $(P, R) = (8, 1), (16, 2), (32, 4)$, and median local coefficients of variation at the three relative window sizes above. Exact boundary handling and numerical constants are specified in the versioned feature implementation identified in Appendix G.

## A.2 Study 1 collection details

Reference eligibility screening excluded one figure-led work. The retained paintings are not matched to individual generated compositions.

NB2 and FLUX deliver 1024-by-1024 images in JPEG and PNG respectively. OAuth delivers PNGs with variable geometry: 424 of its 430 images report low quality and six report medium, all six in artist-free conditions. These delivery differences are retained in the prompt comparison. All three OAuth conditions are randomly ordered within each scene/repetition group. Repetitions are distributed across eight assigned collection batches, which do not denote independent backend sessions.

The service totals are 288 NB2, 288 FLUX and 430 OAuth images. Two Cézanne/OAuth short-scene requests are refused and remain missing. One transient FLUX failure is recovered by an exact-payload technical retry. All successful outputs are retained without selection for stylistic resemblance.

## A.3 Exploratory prompt construction

Let $S$ be the unchanged artist-free scene sentence, beginning "An oil painting on canvas of ...". The by-name condition inserts "by [painter]" immediately after "canvas"; its control is $S$. The explicit-style condition appends "Render this scene in the painting style of [painter]." to $S$, whereas its artist-free control appends "Render this scene in a coherent painting style." The style-plus-aspects method adds the same final sentence to both of these conditions: "Attend to color relationships, the organization of forms, and the handling of painted marks; retain the stated scene." Each method therefore has its own control wording. These contrasts differ from the single sentence addition tested in Study 1.

### A.4 Controlled-study prompt construction

An example detailed named prompt is:

> Create an oil painting on canvas. In the style of Claude Monet. Scene: A pond viewed from close to its edge, with groups of floating lily pads, gaps of open water and reflections of surrounding trees. The water fills the scene; no horizon or prominent person is visible. Render only the painting area, without a surrounding frame, signature, letters or watermark.

The artist-free version removes only the painter-style sentence. The short-scene named version substitutes "A pond with water lilies." for the detailed scene. All 24 descriptions are available in the accompanying prompt inventory.

Study 2 uses six separate fixed descriptions, covering a river and river bend, a village and houses, and fields and a garden. The common opening and closing instructions are retained. The style sentence is absent, generic ("In a traditional landscape-painting style."), Monet or Cézanne. The palette sentence is "Use a muted, low-chroma palette throughout the painting." or "Use a vivid, high-chroma palette throughout the painting." Each of the 48 scene/style/palette combinations has four outputs. The exact six scene texts and complete request inventory accompany the analysis.

## B Complete four-painter distribution summaries

Table 8 reports all 24 cells underlying the main-text ranges. Aliases 1/2 denote the requested `gpt-image-1`/`gpt-image-2` routes; each cell has 64 generated images.

A four-fold sensitivity analysis grouped by assigned generation block gives by-name RBF balanced accuracies .986/.983 for Sisley and .960/.967 for Pissarro under aliases 1/2. These descriptive checks use the original block labels; the later retry images were not generated at those block times. Neither split evaluates transfer to independent generation sessions or capture sources.

## C Paired randomization statistic

Let $A_i, B_i$ be a retained pair with common weight $b_i$, and define $C(v) = \sum_j a_j \|v - x_j\|_2$ and $T(v) = \sum_i b_i(\|v - A_i\|_2 + \|v - B_i\|_2)$. Then

$$c_i = b_i\{2[C(B_i) - C(A_i)] - [T(B_i) - T(A_i)]\}, \qquad \sum_i c_i = \widehat{\mathcal{E}}(X, B) - \widehat{\mathcal{E}}(X, A). \tag{6}$$

Complementary pair swaps produce $\sum_i s_i c_i$, $s_i \in \{-1, 1\}$, exactly, including the within-generated distances. Independent randomized condition positions justify the sign assignments under the stated sharp null. Repeated scene descriptions are not treated as independently sampled scenes for population inference. With $e$ sign draws at least as extreme in absolute value as the observed statistic, including ties, $p = (1 + e)/100{,}000$. All contrasts retain 24 distinct descriptions; the two missing short-scene outputs reduce only the Cézanne detailed/short-scene comparison to 70 pairs.

| Painter | Alias | Prompt | Trace ratio | Linear BA | RBF BA |
|---|---|---|---|---|---|
| Monet | 1 | By name | 0.249 | 0.946 | 0.953 |
| Monet | 1 | Style | 0.230 | 0.970 | 0.961 |
| Monet | 1 | Style + aspects | 0.227 | 0.973 | 0.980 |
| Monet | 2 | By name | 0.304 | 0.954 | 0.947 |
| Monet | 2 | Style | 0.206 | 0.955 | 0.982 |
| Monet | 2 | Style + aspects | 0.209 | 0.973 | 0.979 |
| Sisley | 1 | By name | 0.353 | 0.969 | 0.983 |
| Sisley | 1 | Style | 0.223 | 0.976 | 0.983 |
| Sisley | 1 | Style + aspects | 0.242 | 0.956 | 0.983 |
| Sisley | 2 | By name | 0.314 | 0.961 | 0.970 |
| Sisley | 2 | Style | 0.227 | 0.964 | 0.983 |
| Sisley | 2 | Style + aspects | 0.219 | 0.947 | 0.983 |
| Pissarro | 1 | By name | 0.342 | 0.934 | 0.940 |
| Pissarro | 1 | Style | 0.240 | 0.950 | 0.979 |
| Pissarro | 1 | Style + aspects | 0.240 | 0.946 | 0.967 |
| Pissarro | 2 | By name | 0.376 | 0.945 | 0.944 |
| Pissarro | 2 | Style | 0.267 | 0.946 | 0.963 |
| Pissarro | 2 | Style + aspects | 0.256 | 0.939 | 0.955 |
| Cézanne | 1 | By name | 0.309 | 0.951 | 0.978 |
| Cézanne | 1 | Style | 0.255 | 0.942 | 0.965 |
| Cézanne | 1 | Style + aspects | 0.262 | 0.942 | 0.948 |
| Cézanne | 2 | By name | 0.294 | 0.964 | 0.973 |
| Cézanne | 2 | Style | 0.254 | 0.913 | 0.942 |
| Cézanne | 2 | Style + aspects | 0.249 | 0.942 | 0.968 |

**Table 8:** All 24 four-painter cells in the full 31-feature space. Ratios use separately centered sample traces; both classifier scores pool out-of-fold predictions under the fixed 16-scene split. The entries are descriptive estimates without uncertainty intervals.

# D Supporting Study 1 distribution diagnostics

## D.1 Matched reference-neighborhood coverage

For each painter, each of 100 fixed draws splits the references into disjoint anchor and real-query groups within content class. The groups have 18 works each for Monet (water/built/land 10/2/6) and 15 each for Cézanne (1/5/9); unmatched odd works are omitted in that draw. Generated queries are sampled without replacement with the same class counts. Every anchor's radius is its distance to the $k$th nearest other anchor in the fixed 31-feature, primary-512 space. Coverage is the fraction of anchor balls containing at least one query. Real and generated queries share each draw's anchors and radii. The draws reuse the panel; medians and draw ranges do not estimate uncertainty across independently sampled painting populations.

| Painter | $k$ | NB2 named | FLUX named | OAuth named | Real |
|---|---|---|---|---|---|
| Monet | 1 | .222 | .444 | .278 | .500 |
| | 3 | .500 | .833 | .500 | .944 |
| | 5 | .667 | .944 | .722 | 1.000 |
| Cézanne | 1 | .133 | .467 | .333 | .533 |
| | 3 | .400 | .800 | .533 | .933 |
| | 5 | .667 | 1.000 | .733 | 1.000 |

**Table 9:** Median matched-anchor coverage across 100 finite-panel draws. Neighborhood size changes the numerical comparison; saturation at $k = 5$ for FLUX/Cézanne does not establish matching distributions. The accompanying numerical tables include artist-free conditions and all draw memberships.

### D.2 Alignment controls

Equation 4 uses equal broad-class masses. Sparse reference strata therefore receive one-third weight, giving reference effective sample sizes of about 24.0 for Monet and 18.8 for Cézanne; the nominal counts are 38 and 32. The weighting changes the contribution of existing works; it does not add independent reference information. For all 216 matched artist-free pairs across painter labels, request payloads are verified identical. Their separate finite collections have V-energies .079, .102 and .077 for NB2, FLUX and OAuth. These values illustrate collection and finite-sample variation, without defining an equivalence margin or a significance threshold for alignment.

### D.3 Processing, representation and reference sensitivities

We examine three processing pipelines: the primary 512-pixel pipeline, a 256-pixel version and common JPEG encoding at quality 90 with 4:2:0 subsampling at 512 pixels. Each uses its own development-only scaler. Four feature views cross these pipelines: all 31 coordinates; 28 coordinates after dropping three derived multiscale summaries; 31 coordinates divided by the square root of their family size; and the 19 color/spatial coordinates with texture removed. The family-rescaled squared distance sums mean squared differences within each family; it does not equalize observed family variance or perceptual importance. This gives 12 views per service–painter comparison.

Reference diagnostics delete one work, holding-collection membership group or content class at a time. Collection groups use exact recorded sets of holding- collection IDs and do not identify photographic capture workflows. Work and collection-group deletions preserve the original content-class masses while reweighting the remaining works within each class; class deletion renormalizes the surviving class masses. A separate analysis refits the scaler after omitting each development painter.

### D.4 Projection and domain detection

For visualization, we fit one common principal-component analysis (PCA) per painter, assigning half the weight to references and sharing half equally across the seven generated cells, with content weighting within each. All cells use that painter's common axes. Full-space linear and radial-basis-function (RBF) kernel-ridge classifiers use six folds holding out entire generated descriptions and disjoint original works, without tuning or PCA inputs. This grouping respects the repeated-prompt structure (Roberts et al., 2017). Cross-service evaluation trains on one service and tests on another using the same held-out description/work scheme. We report balanced accuracy at a fixed zero decision threshold and mean within-fold area under the ROC curve (AUC) separately; the former also depends on calibration.

The PCA views show overlapping clouds with changes in location and concentration (Figure 6). The first two components retain 48.4% of weighted variation for Monet and 46.0% for Cézanne. With more than half the variation omitted, these projections provide only a partial view of the distributions.

In the full feature space, within-service RBF balanced accuracy ranges from .893 to .987 across the 14 original/generated comparisons. However, all 576 NB2/FLUX images are square and all 70 references are nonsquare: a rule using only square geometry separates these domains perfectly. Capture workflows for the reference reproductions are also unresolved. High classification accuracy therefore establishes differences between the recorded image domains, not a validated stylistic distinction.

Cross-service behavior further limits detector interpretation. At the fixed threshold, naming lowers balanced accuracy in every directed service/painter comparison for both kernels. Named-

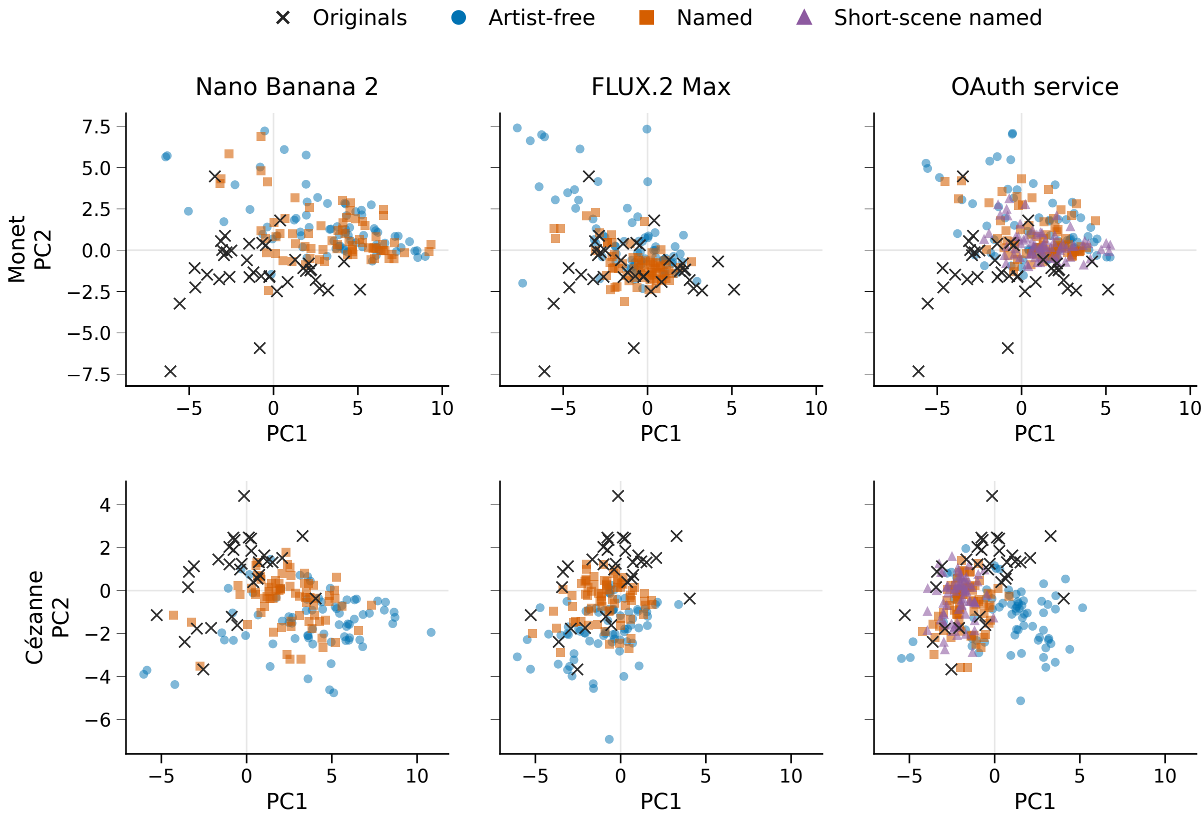


**Figure 6:** Original and generated feature distributions in common painter-specific PCA coordinates. Every Study 1 image is shown; references are repeated across service panels for comparison. Axes and limits are shared within each painter, but the two painters have separate PCA fits. The short-scene named condition is available only for OAuth. Projections show overlap and concentration while retaining less than half of total weighted variation.

condition ranges are .410–.874 for the linear kernel and .447–.704 for RBF. This need not mean uniformly weaker ranking discrimination. For Monet with training on FLUX and testing on OAuth, linear balanced accuracy falls from .904 to .874 while mean within-fold AUC rises from .953 to .968. The observed threshold transfer includes calibration changes; below-chance scores are not evidence that generated images have become more authentic.

### D.5 Classifier specification

For $d = 31$ standardized coordinates, the fixed kernels are $K_{\mathrm{L}}(u, v) = u^\top v/d + 1$ and $K_{\mathrm{R}}(u, v) = \exp(-\|u - v\|_2^2/d) + 1$. The classifiers minimize weighted squared error with a kernel penalty of .01, using equal total training weight for original and generated labels and the reference content proportions within each. Training and evaluation memberships are fixed. Sparse reference classes mean not every test fold contains all three classes; pooled balanced accuracy uses the complete held-out set. For cross-service evaluation, AUC is computed within each fold and averaged across six folds; margins from differently trained classifiers are not pooled. Complete kernels, memberships and scores are included with the analysis code and numeric tables.

## E Color experiment: estimation and acquisition

For fixed scene $j = 1, \ldots, J$, repetition $r = 1, \ldots, R$ and arm $a$, let $D_{jra} = z_{jra,+} - z_{jra,-}$ and $K_{jr} = (D_{jrM} - D_{jrG}, D_{jrC} - D_{jrG})^\top$, with $J = 6$ and $R = 4$. Writing $\bar{K}_j = R^{-1} \sum_r K_{jr}$ and

$S_j$ for their within-scene sample covariance, the estimates and covariance are

$$\hat{\kappa} = \frac{1}{J}\sum_j \bar{K}_j, \qquad \widehat{\mathrm{Cov}}(\hat{\kappa}) = \sum_j \frac{S_j}{J^2 R}. \tag{7}$$

This retains the off-diagonal covariance from shared generic controls. For endpoint $p$, let $v_{jp} = S_{j,pp}/(J^2R)$. The approximate degrees of freedom are

$$\nu_p = \frac{(\sum_j v_{jp})^2}{\sum_j v_{jp}^2/(R-1)}. \tag{8}$$

The two simultaneous intervals are $\hat{\kappa}_p \pm t_{1-.05/(2\cdot 2),\nu_p}\sqrt{\sum_j v_{jp}}$. Two-sided $t$ $p$-values receive Holm adjustment within this two-endpoint family. The scenes are fixed; between-scene variation of mean responses is not added as a random-scene variance component. Estimated scene means retain generation noise. Request-order randomization does not establish the assumed independence and stability of repeat errors.

## E.1 Repeat-block variation

Figure 7 is a post-result display of all 24 observed interactions for each painter. Monet has 17 negative block contrasts, although all six scene means are negative. The garden scene contributes 25.7% of Monet's estimated interaction variance; the fields scene contributes 44.5% of Cézanne's. These shares are $v_{jp}/\sum_j v_{jp}$ under the unchanged stratified estimator. They describe the concentration of estimated uncertainty within the six fixed scenes. No blocks are removed, and the display does not establish independence or stability across service collections.

## E.2 Prospective calibration and collection history

A prospective precision simulation used centered historical OAuth within-scene residuals with finite-repeat variance correction. The generic arm's noise was proxied by artist-free residuals, without validation that the two distributions match. Each of three independent-noise scenarios used 5,000 trials for each specified interaction setting. Besides the empirical proxy, a second scenario multiplied generic noise by 1.5. A heteroskedastic normal scenario applied scene standard-deviation multipliers from .7 to 1.8 to arm-specific baseline noise, then muted/vivid factors of .8/1.4, a generic factor of 1.5 and a Cézanne factor of 1.25. Table 10 reports the global-null rows. Null-family error over the partial-null settings ranges from .0336 to .0428. The hypothetical .25/.5/1-IQR interactions assess precision under specified effect sizes; they are not perceptual thresholds or post-hoc power estimates. These simulations support the procedure under complete-availability, independent proxy errors; they do not establish realized-service independence, tail behavior or coverage.

| Noise scenario | Holm family rejection | Simultaneous interval coverage |
|---|---|---|
| Empirical proxy | .0382 [.0332, .0439] | .9618 [.9561, .9668] |
| Generic noise $\times$1.5 | .0330 [.0284, .0383] | .9670 [.9617, .9716] |
| Heteroskedastic normal | .0404 [.0353, .0462] | .9596 [.9538, .9647] |

**Table 10:** Prospective global-null calibration in 5,000 trials per scenario. Brackets are 95% Wilson Monte Carlo intervals for simulation proportions, not confidence intervals for experimental effects. Each row uses independent simulated repeat errors and complete availability.

An earlier allocation was stopped after 49 images and one complete HTTP 503 response, leaving 142 assigned slots unstarted. A transport correction and a replacement collection using

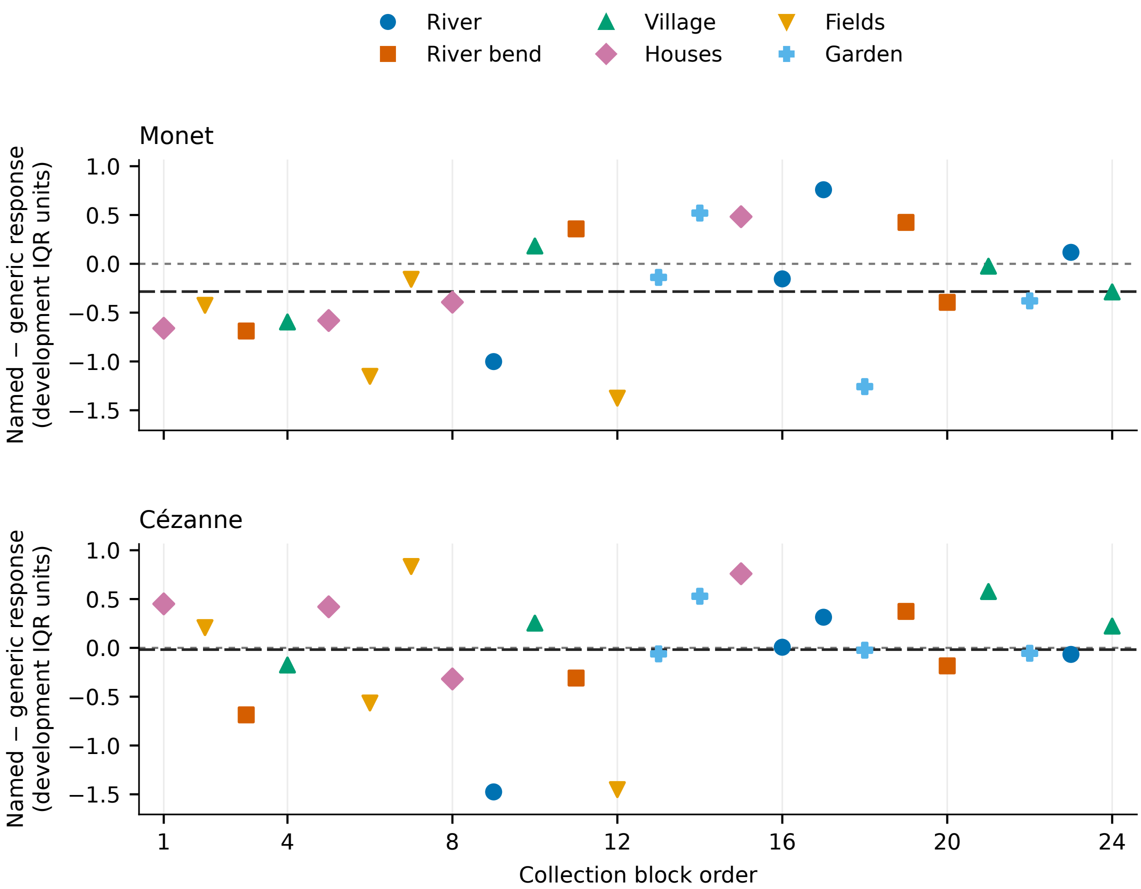


**Figure 7:** All repeat-block named-minus-generic chroma-response interactions in Study 2. Each point combines the named and shared generic vivid/muted outcomes from one eight-request block. Colors and symbols identify the six scenes; the horizontal axis gives collection-block order, which matched the fixed dispatch schedule. Both panels retain all 24 blocks on a common vertical scale. Reference lines mark zero and the pooled interaction estimate with short and long dashes, respectively. This descriptive display adds no significance test.

the same fixed prompts and order were specified before visual inspection or feature measurement of the successful images. The 192-image replacement is the sole Study 2 inferential cohort. The 49 earlier images were measured after replacement collection ended and remain ancillary, without pooling or filling missing slots. All 192 replacement requests succeeded, with no failures, retries or missing measurements.

# F Designed chroma mixtures and the reference panels

The named arms' equal-polarity mixtures differ from the reference panels in both mean chroma and the allocation of probability mass (Table 11). Most reference paintings lie inside the generated central range, while most generated outputs lie outside the reference central range. Broad range inclusion therefore does not establish a matched distribution.

Equal broad-class reference weighting gives named-mixture Wasserstein distances of .603 for Monet and .759 for Cézanne, compared with 1.071 and .787 for the same generic-control mixture. Naming lowers this distance in both cases, with directions preserved across all three pipelines and both reference weightings. The comparison shows closer chroma distributions under naming for these designed mixtures. The two extreme instructions do not establish continuous coverage of the intervening range.

| | Mean chroma | | Mass inside central range | | |
|---|---|---|---|---|---|
| Painter | Original | Generated | Gen. in orig. | Orig. in gen. | $W_1$ |
| Monet | .369 | .533 | 37.5% | 89.5% | .620 |
| Cézanne | .812 | 1.075 | 16.7% | 93.8% | .716 |

**Table 11:** Chroma distributions in Study 2 relative to the empirical reference panel. Generated values use each painter's 48 named outputs with equal class and polarity mass. Central ranges are empirical 10th–90th percentiles, not confidence intervals. Means and Wasserstein-1 distances use development-IQR units. The forced muted/vivid mixture is not an unprompted sampling distribution.

# G Analysis files and reproduction

The scientific source, fixed analysis memberships and numerical reports are versioned together in snapshot `28a9eb6`. The feature specification is implemented in `src/latent_art_bench/painter_feature_generation_v2/features.py`; the common scaler is recorded in `data/manifests/painter_feature_generation_v2/pfg2-method-20260905/scaler.json`. The current manuscript build scripts are `paper/make_figures.py` and `paper/replay_palette.py`, supplied with this revision rather than the earlier scientific snapshot. The principal report directories are:

- Four-painter exploration and controls: `reports/painter_distribution_exploration_v1/`; `reports/painter_distribution_study_v1/pdsv1-diagnostics-20260906/`; `reports/painter_prompt_retry_v1/ppr1-two-refusals-20260906-r2/`.
- Study 1 and its subsequent diagnostics: `reports/painter_distribution_study_v1/`; `reports/painter_distribution_revision_v1/`.
- Retrieval and Study 2: `reports/painter_responsiveness_v2/`. The primary color collection is `prv2-oauth-recovery-20260908`.
- Corrected descriptive quantiles: `reports/painter_responsiveness_quantiles_v1/`.

The build guide `paper/README.md` maps these reports to their protocols, source files and reproduction commands. Six manuscript figures extract saved table values and coordinates; the repeat-block figure derives its contrasts from retained chroma measurements and the fixed schedule. The command `make palette-check` recomputes Study 2's primary intervals using the unchanged inference implementation and checks them against the saved results. It requires only committed compact inputs. Full computational replay also verifies response hashes against the separately retained archive. Neither command regenerates images.

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